Практика 1. Отчёт. Выполнил: Болдырев Денис АИ-32

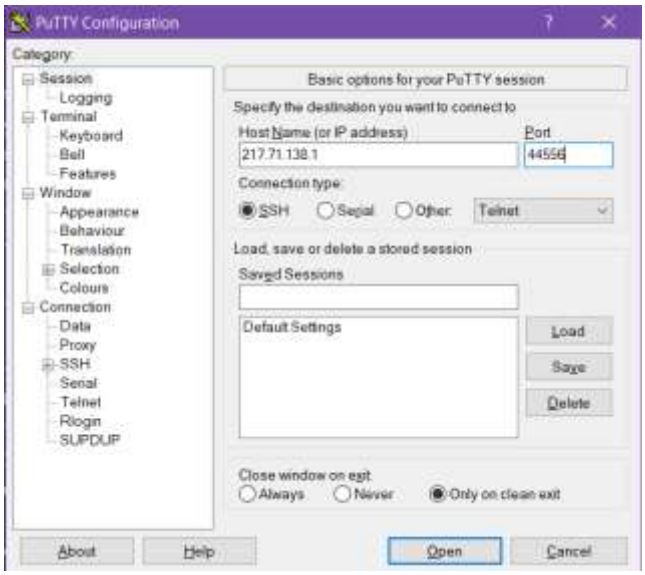
Подключение к серверу используя программу ssh:

```
C:\Users\mega1>ssh eltex-pg1-v15@217.71.138.1 -p 44556
eltex-pg1-v15@217.71.138.1's password:
Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Mar  5 17:50:00 2026 from 217.71.131.234
eltex-pg1-v15@eltex-2026-spring:~$
```

Подключение к серверу используя программу putty:




```
login as: eltex-pg1-v15
eltex-pg1-v15@217.71.138.1's password:
Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (
2025-09-20) x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
Last login: Thu Mar  5 17:50:32 2026 from 217.71.131.234
eltex-pg1-v15@eltex-2026-spring:~$
```
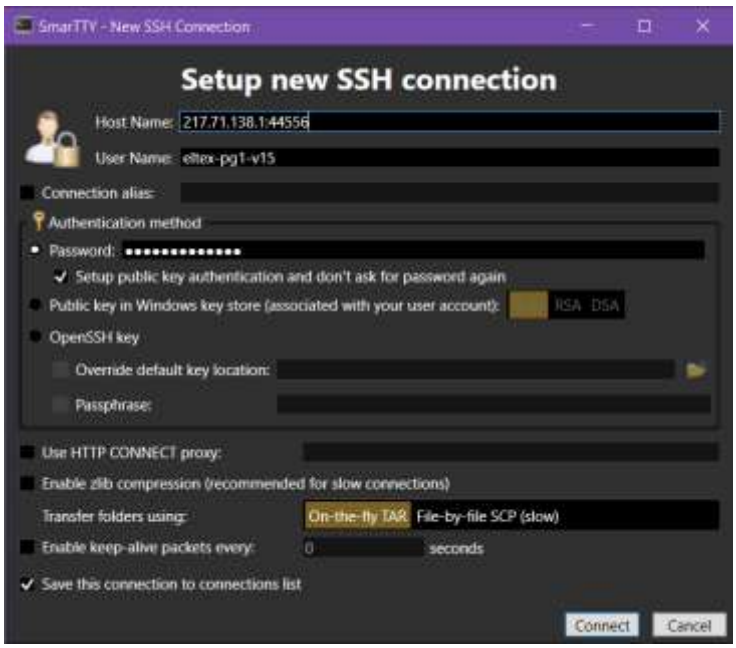
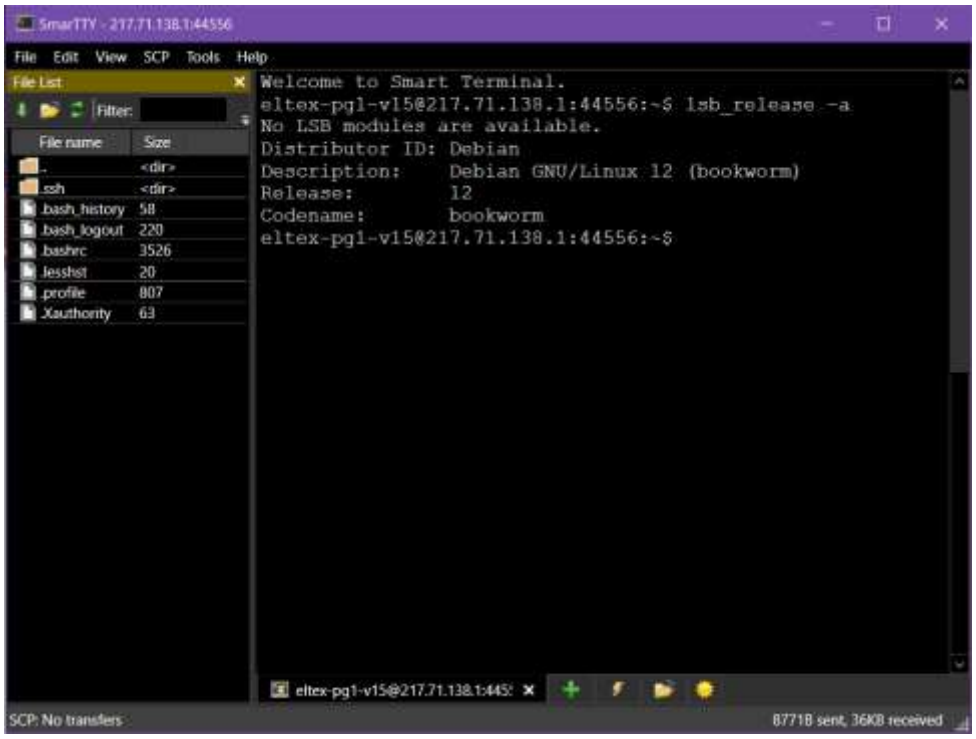

Подключение к серверу используя программу smartty:


SmarTTY - New SSH Connection
Setup new SSH connection
Host Name: 217.71.138.1:44556
User Name: eltex-pg1-v15
Connection alias:
Authentication method
Password:
Setup public key authentication and don't ask for password again
Public key in Windows key store (associated with your user account):
RSA DSA
OpenSSH key
Override default key location:
Passphrase:
Use HTTP CONNECT proxy:
Enable zlib compression (recommended for slow connections)
Transfer folders using:
On-the-fly TAR
File-by-file SCP (slow)
Enable keep-alive packets every:
0
seconds
Save this connection to connections list
Connect
Cancel


SmarTTY - 217.71.138.1:44556
File Edit View SCP Tools Help
File List
Filter:
File name Size
.. <dir>
.ssh <dir>
.bash_history 58
.bash_logout 220
.bashrc 3526
.lesshst 20
.profile 807
.Xauthority 63
Welcome to Smart Terminal.
eltex-pg1-v15@217.71.138.1:44556:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Debian
Description: Debian GNU/Linux 12 (bookworm)
Release: 12
Codename: bookworm
eltex-pg1-v15@217.71.138.1:44556:~$
eltex-pg1-v15@217.71.138.1:445
SCP: No transfers
8771B sent, 36KB received

Запуск новой сессии утилиты screen, с именем по умолчанию, и просмотр списка подключённых пользователей командой w:

```
eltex-pg1-v15@eltex-2026-spring: ~
eltex-pg pts/35   :pts/20:S.0      18:38    1:11   0.00s   ?    watch /usr/bin/vmstat
eltex-pg pts/44   :pts/10:S.0      18:39    6.00s  0.14s  0.14s w
eltex-pg pts/46   :pts/3:S.0       18:23    2:31   0.03s  0.01s SCREEN -S eltex-pg1-v12_part2
eltex-pg pts/47   188.162.13.15    17:45   14:39   0.03s   ?    screen -S whoami_part2
eltex-pg pts/48   217.71.139.234   17:44   52:45   0.01s  0.01s -bash
eltex-pg pts/50   :pts/14:S.4      18:24   45:18   0.01s  0.01s /usr/bin/bash
eltex-pg pts/55   :pts/60:S.0      17:54   28:13   0.00s   ?    screen -r top
eltex-pg pts/56   :pts/16:S.0      18:24   13.00s  0.08s  0.04s SCREEN -S 'whoami'_part2
eltex-pg pts/59   :pts/9:S.0       18:39    6.00s  0.08s  0.08s vi report_part1.txt
eltex-pg pts/60   :pts/38:S.0      17:50   28:13   0.00s   ?    screen -S top
eltex-pg pts/61   217.71.131.234   17:50   47:30   0.01s  0.01s -bash
eltex-pg pts/62   217.71.139.234   17:59    7.00s  0.03s   ?    screen -r pts-62.eltex-2026-spring
eltex-pg pts/53   91.78.233.73     17:52    1:43   0.01s   ?    screen
eltex-pg pts/70   :pts/14:S.5      18:24   43:59   1.03s  1.03s watch /usr/bin/vmstat
eltex-pg pts/72   :pts/21:S.0      18:33   15.00s  0.00s   ?    script -q .log
eltex-pg pts/73   217.71.131.234   17:55   23.00s  0.03s  0.03s -bash
tech     pts/78   217.71.139.234   18:34   11.00s  0.03s  0.03s sshd: tech [priv]
eltex-pg pts/79   217.71.131.234   17:56    5:51   0.00s   ?    -bash
eltex-pg pts/82   :pts/53:S.0      17:56    1:46   0.00s   ?    screen -S top
eltex-pg pts/83   :pts/41:S.0      18:35   38.00s  0.00s   ?    script -q .log
eltex-pg pts/84   :pts/55:S.0      17:58   37:01   0.00s  0.01s SCREEN -S top -d -m
eltex-pg pts/88   :pts/14:S.6      18:24   40:55   0.97s  0.97s watch usr/bin/vmstat
eltex-pg pts/90   :pts/14:S.7      18:24   39:43   0.96s  0.96s watch /usr/bin/vmstat
eltex-pg pts/100  :pts/14:S.8      18:24   38:07   0.00s   ?    /usr/bin/bash
eltex-pg pts/116  217.71.131.234   18:06    1:03   0.02s  0.02s -bash
eltex-pg pts/121  :pts/82:S.0      18:10    1:46   0.00s   ?    SCREEN -S top
eltex-pg pts/122  91.78.233.217    18:10    7:43   0.06s  0.06s -bash
eltex-pg pts/120  :pts/53:S.1      18:11    2:29   0.00s   ?    /usr/bin/bash
eltex-pg pts/130  :pts/125:S.0     18:14    7:18   0.00s   ?    SCREEN -S top -d -m
eltex-pg1-v15@eltex-2026-spring:~$
```

Запуск отсоединенной сессии утилиты screen, с именем top и запуском команды top:

Получение списка сессий, созданных утилитой screen:

Подсоединение к сессии top:

```
eltex-pg1-v15@eltex-2026-spring: ~
eltex-pg1-v15@eltex-2026-spring:~$ screen
[detached from 95770.pts-10.eltex-2026-spring]
eltex-pg1-v15@eltex-2026-spring:~$ [detached from 95770.pts-10.eltex-2026-spring]
eltex-pg1-v15@eltex-2026-spring:~$ screen -S top -d -m top
eltex-pg1-v15@eltex-2026-spring:~$ screen -ls
There are screens on:
        97280.top       (03/05/2026 06:40:35 PM)        (Detached)
        95770.pts-10.eltex-2026-spring  (03/05/2026 06:39:29 PM)        (Detached)
2 Sockets in /run/screen/S-eltex-pg1-v15.
eltex-pg1-v15@eltex-2026-spring:~$ screen -r top
[detached from 97280.top]
eltex-pg1-v15@eltex-2026-spring:~$
```

```
eltex-pg1-v15@eltex-2026-spring: ~
top - 18:41:13 up  7:18, 50 users,  load average: 0.02, 0.04, 0.07
Tasks: 419 total,   1 running, 414 sleeping,   4 stopped,   0 zombie
%Cpu(s):  0.4 us,  0.7 sy,  0.0 ni, 98.5 id,  0.1 wa,  0.0 hi,  0.3 si,  0.0 st
MiB Mem :   3914.7 total,   2253.7 free,    827.9 used,   1074.4 buff/cache
MiB Swap:    976.0 total,    976.0 free,      0.0 used.   3086.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  35327 eltex-p+  20   0   11640   5548   3416 S   0.3   0.1   0:06.86 top
  35427 eltex-p+  20   0   11640   5620   3476 S   0.3   0.1   0:05.45 top
  37911 eltex-p+  20   0   11640   5504   3356 S   0.3   0.1   0:04.74 top
  38887 eltex-p+  20   0   11640   5568   3420 S   0.3   0.1   0:04.51 top
  45057 eltex-p+  20   0    6724   3364   2416 S   0.3   0.1   0:00.92 watch
  45707 eltex-p+  20   0   11780   5500   3356 S   0.3   0.1   0:04.34 top
  53191 eltex-p+  20   0   11632   5564   3416 S   0.3   0.1   0:04.73 top
  56313 eltex-p+  20   0   11640   5552   3408 S   0.3   0.1   0:02.95 top
  61187 eltex-p+  20   0   11640   5544   3400 S   0.3   0.1   0:02.61 top
  66129 root      20   0       0      0      0 I   0.3   0.0   0:06.37 kworker/u8:5-events_unbound
  79876 eltex-p+  20   0    6184   2892   2472 S   0.3   0.1   0:00.15 watch
  92769 eltex-p+  20   0   11644   5572   3420 S   0.3   0.1   0:00.40 top
  97281 eltex-p+  20   0   11640   5548   3400 R   0.3   0.1   0:00.07 top
      1 root      20   0  168644  13052   9204 S   0.0   0.3   0:01.61 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
```

Создание ещё одного окна в сессии top и выполнение команды «watch /usr/bin/vmsstat»:

```
eltex-pg1-v15@eltex-2026-spring: ~
Every 2.0s: /usr/bin/vmstat                          eltex-2026-spring: Thu Mar  5 18:44:37 2026

procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 3  0      0 2300200  73188 1027448    0    0     4    18   79  170  0  0 99  1  0
```

Получение списка окон с помощью комбинации клавиш «ctrl-a + “»:

```
eltex-pg1-v15@eltex-2026-spring: ~
Num Name                                                                    Flags

  0 top                                                                        $
  1 bash                                                                       $
```

Переименование второго окна как «vmstat»:

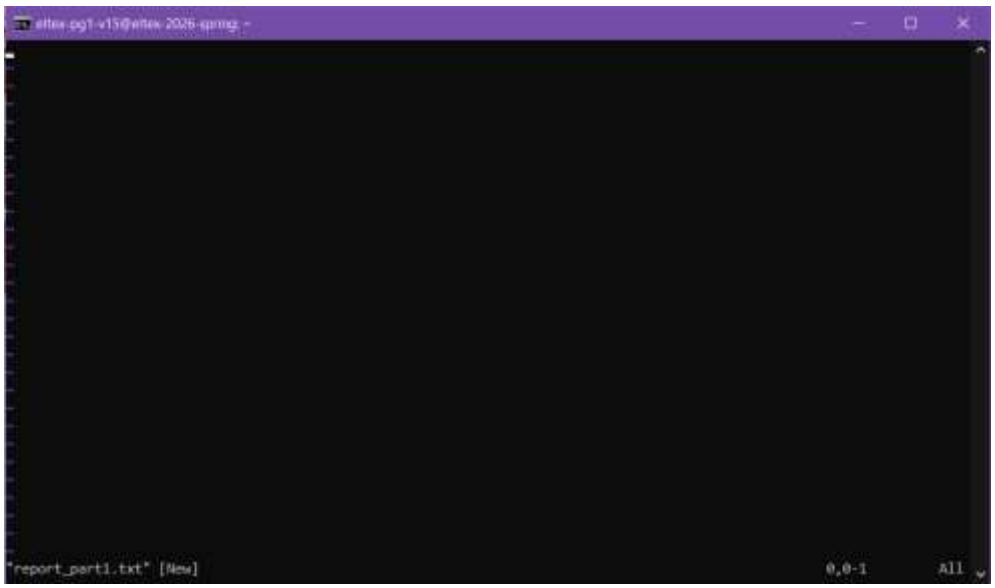
Использована комбинация клавиш «ctrl-a + A».

```
Set window's title to: vmstat
```

```
Выбрать eltex-pg1-v15@eltex-2026-spring: ~
Num Name                                                                    Flags

  0 top                                                                        $
  1 vmstat                                                                     $
```

Затем было выбрано окно top и для отсоединения от сессии использована комбинация клавиш «ctrl-a + d».

Запуск новой сессии и запуск в окне команды «vi report_part1.txt»:



Отсоединение от сессии и получение списка сессий:

```
eltex-pg1-v15@eltex-2026-spring:~$ eltex-pg1-v15@eltex-2026-spring:~$ screen -S new_session
[detached from 113507.new_session]
eltex-pg1-v15@eltex-2026-spring:~$ [detached from 113507.new_session]
eltex-pg1-v15@eltex-2026-spring:~$ screen -ls
There are screens on:
        113507.new_session      (03/05/2026 06:51:43 PM)        (Detached)
        97280.top       (03/05/2026 06:40:34 PM)        (Detached)
        95770.pts-10.eltex-2026-spring  (03/05/2026 06:39:28 PM)        (Detached)
3 Sockets in /run/screen/S-eltex-pg1-v15.
eltex-pg1-v15@eltex-2026-spring:~$
```

Отключился от сервера с помощью команды logout.

Подключился заново и просмотрел список сессий screen:

```
eltex-pg1-v15@eltex-2026-spring: ~
eltex-pg1-v15@eltex-2026-spring:~$ [detached from 113507.new_session]
eltex-pg1-v15@eltex-2026-spring:~$ screen -ls
There are screens on:
        113507.new_session      (03/05/2026 06:51:43 PM)        (Detached)
        97280.top       (03/05/2026 06:40:34 PM)        (Detached)
        95770.pts-10.eltex-2026-spring  (03/05/2026 06:39:28 PM)        (Detached)
3 Sockets in /run/screen/S-eltex-pg1-v15.
eltex-pg1-v15@eltex-2026-spring:~$ logout
There are stopped jobs.
eltex-pg1-v15@eltex-2026-spring:~$ logout
Connection to 217.71.138.1 closed.

C:\Users\mega1>ssh eltex-pg1-v15@217.71.138.1 -p 44556
eltex-pg1-v15@217.71.138.1's password:
Linux eltex-2026-spring 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
Last login: Thu Mar  5 18:09:06 2026 from 5.141.195.65
eltex-pg1-v15@eltex-2026-spring:~$ screen -ls
There are screens on:
        113507.new_session      (03/05/2026 06:51:44 PM)        (Detached)
        97280.top       (03/05/2026 06:40:35 PM)        (Detached)
        95770.pts-10.eltex-2026-spring  (03/05/2026 06:39:29 PM)        (Detached)
3 Sockets in /run/screen/S-eltex-pg1-v15.
eltex-pg1-v15@eltex-2026-spring:~$
```

Открытые сессии были сохранены.

Запуск редактора, запись необходимо текста, затем сохранение и выход из редактора с помощью «:wq»:

```
eltex-pg1-v15@eltex-2026-spring: ~
eltex-pg1-v15@eltex-2026-spring:~$ vi repot_part1.txt

Болдырев Денис Андреевич часть 1 модуля 1 успешно завершена 05.03.2026 18:58
~
~
~
```

eltex-pg1-v15@eltex-2026-spring: ~

```
eltex-pg1-v15@eltex-2026-spring:~$ HISTTIMEFORMAT="%d/%m/%y %T "
eltex-pg1-v15@eltex-2026-spring:~$ date
Thu Mar 12 03:30:35 PM +07 2026
eltex-pg1-v15@eltex-2026-spring:~$ history 5
  378  12/03/26 15:18:11 ls -d ~ .*
  379  12/03/26 15:30:05 $HISTTIMEFORMAT="%d/%m/%y %T "
  380  12/03/26 15:30:26 HISTTIMEFORMAT="%d/%m/%y %T "
  381  12/03/26 15:30:35 date
  382  12/03/26 15:30:41 history 5
eltex-pg1-v15@eltex-2026-spring:~$ tail 5 .bash_history
tail: cannot open '5' for reading: No such file or directory
==> .bash_history <==
man history
history 12
man history
echo $HISTTIMEFORMAT
HISTTIMEFORMAT
man bash
man history
screen -r 'whoami'_part2
tail $HISTFILE
ls -d ~ .*
eltex-pg1-v15@eltex-2026-spring:~$ PROMPT_COMMAND="history -a; $PROMPT_COMMAND"
eltex-pg1-v15@eltex-2026-spring:~$ history 5
  381  12/03/26 15:30:35 date
  382  12/03/26 15:30:41 history 5
  383  12/03/26 17:56:26 tail 5 .bash_history
  384  12/03/26 17:57:32 PROMPT_COMMAND="history -a; $PROMPT_COMMAND"
  385  12/03/26 17:57:38 history 5
eltex-pg1-v15@eltex-2026-spring:~$ tail 5 .bash_history
tail: cannot open '5' for reading: No such file or directory
==> .bash_history <==
#1773304235
date
#1773304241
history 5
#1773312986
tail 5 .bash_history
#1773313052
PROMPT_COMMAND="history -a; $PROMPT_COMMAND"
#1773313058
history 5
eltex-pg1-v15@eltex-2026-spring:~$
```

```
eltex-pg1-v15@eltex-2026-spring:~$ DATE="%y %m %d %T"
eltex-pg1-v15@eltex-2026-spring:~$ echo %DATE
%DATE
eltex-pg1-v15@eltex-2026-spring:~$ echo $DATE
%y %m %d %T
eltex-pg1-v15@eltex-2026-spring:~$
eltex-pg1-v15@eltex-2026-spring:~$ DATE='%y %m %d %T'
eltex-pg1-v15@eltex-2026-spring:~$ echo $DATE
%y %m %d %T
eltex-pg1-v15@eltex-2026-spring:~$ DATE=%y %m %d %T
bash: fg: %m: no such job
eltex-pg1-v15@eltex-2026-spring:~$ DATE=$(date +"%d %m %y)
>
> c^C
eltex-pg1-v15@eltex-2026-spring:~$ DATE=$(date +"%d %m %y)
>
> c^C
eltex-pg1-v15@eltex-2026-spring:~$ DATE=$(date +"%d %m %y")
eltex-pg1-v15@eltex-2026-spring:~$ echo $date

eltex-pg1-v15@eltex-2026-spring:~$ echo $DATE
12 03 26
eltex-pg1-v15@eltex-2026-spring:~$ TIME=$(date +"%T")
eltex-pg1-v15@eltex-2026-spring:~$ echo $TIME
18:11:56
eltex-pg1-v15@eltex-2026-spring:~$ DATE_TIME="$DATE $TIME"
eltex-pg1-v15@eltex-2026-spring:~$ echo $DATE_TIME
12 03 26 18:11:56
```